Transfer Learning through Federated Reinforcement Learning

Machine Learning Paradigms Exam

Francesco Refolli

The Multi-Agent Simulation Setting

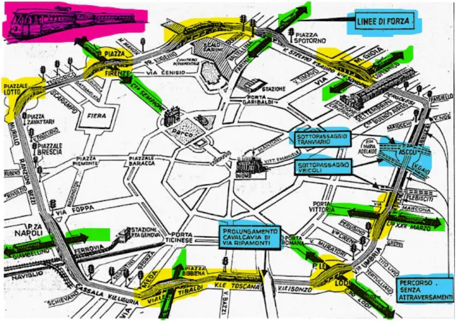


Figure: Ring-Road of a Metropolitan City [5]

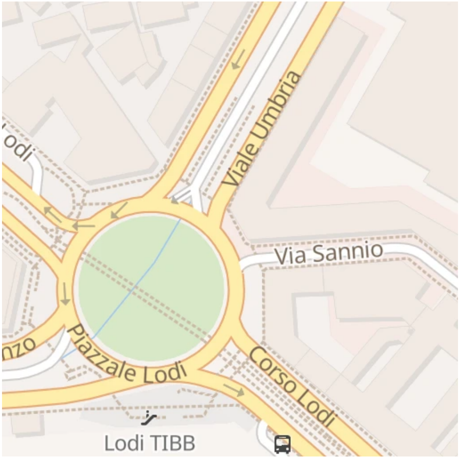


Figure: 4-Way Double-Lane Traffic Light Regulated Intersection [1]

The Multi-Agent Reinforcement Learning Story So Far

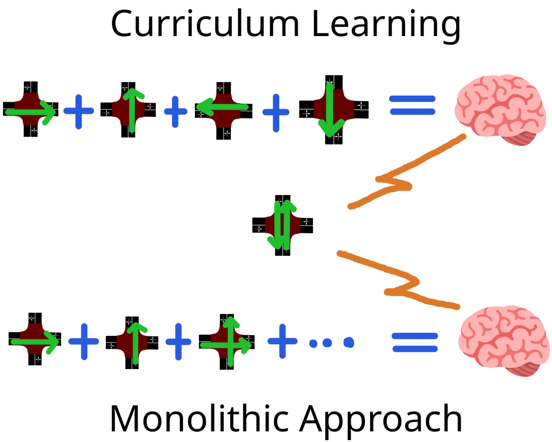


Figure: Comparison of Approaches for Rearching Demand Coverage[2]

The Multi-Agent Reinforcement Learning Story So Far

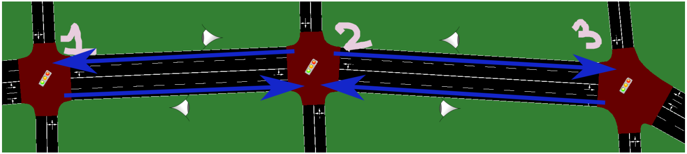


Figure: Agents sharing Observation [2]

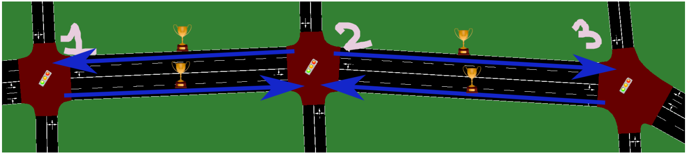
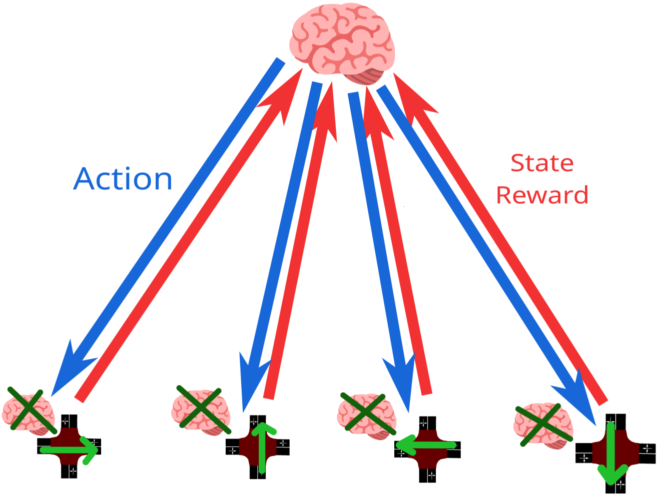
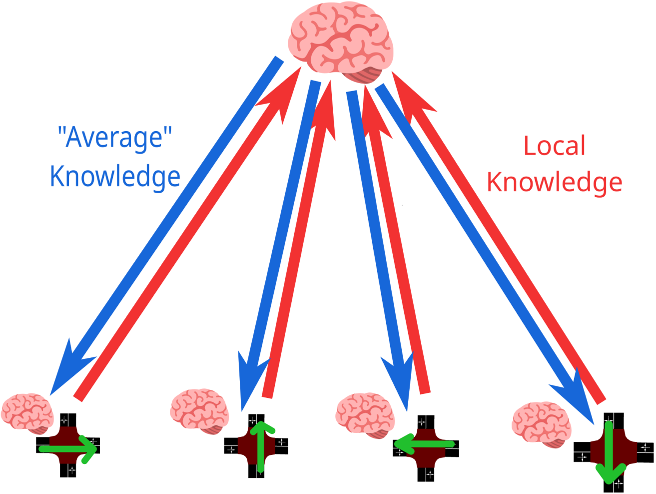


Figure: Agents sharing Rewards [2]

Centralized Reinforcement Learning

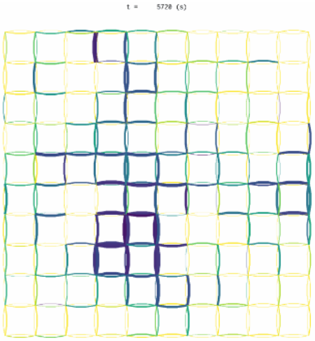


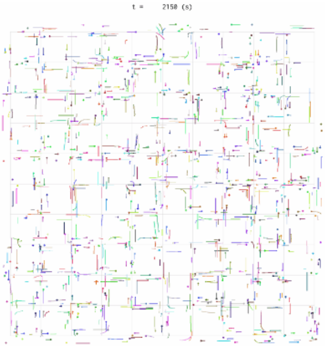
Federated Reinforcement Learning



The Simulator: UXsim

UXSim [3] is a Free and Open-Source *macroscopic* and *mesoscopic* network traffic flow simulator. It's tailored to simulate private-mobility simulations. It's **"Lightweight"**, **Library-Based** and **Written in Python**.





The Agent Model

- The Agent can change Traffic Light Phase every 10 seconds
- Each one of the two *phases* gives *right of way* to one of the two flows
- The Agent can see the density of vehicles in each one of the roads (input/output)
- The Agent is rewarded based on vehicle queues' length, specifically on the amount vehicles en/dequeuing

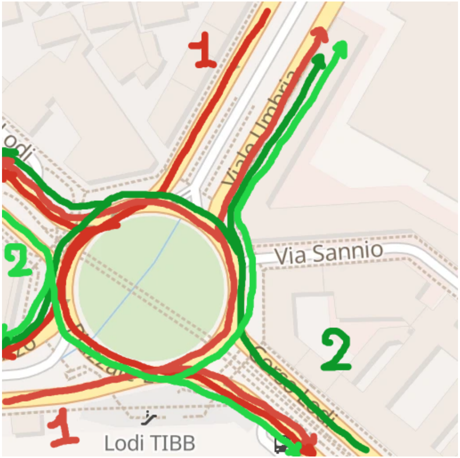


Figure: 4-Way Double-Lane Traffic Light Regulated Intersection

The Traffic Model

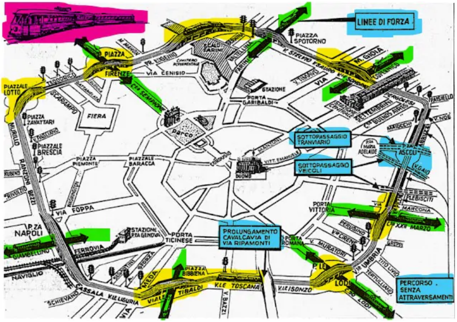


Figure: Ring-Road of a Metropolitan City [5]

- Two-Lanes per Road Link
- Road Length is 1000 m = 1 km
- Free Flow Speed is 20 m/s $\approx$ 72 km/h [4]
- Traffic Jam Density is set to $\rho = 0.2 \approx 40$ veh/2lanes/1km [4]
- No Platoons allowed, as in the Real-World (at least for now ...)
- No simulated Pedestrian Crossing, nor Public Transit (of any kind)

The Demand Model / Training Scenarios

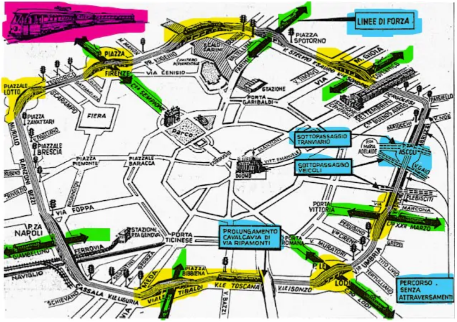


Figure: Ring-Road of a Metropolitan City [5]

- All Mid (**Piazzale Fratelli Zavattari**)
- Peak East to West (**Piazzale Loreto**)
- Peak West to East (**Piazza Simone Bolivar**)
- Peak North to South (**Piazza Firenze**)
- Peak South to North (**Piazzale Lodi**)

The Demand Model / Evaluation Scenarios

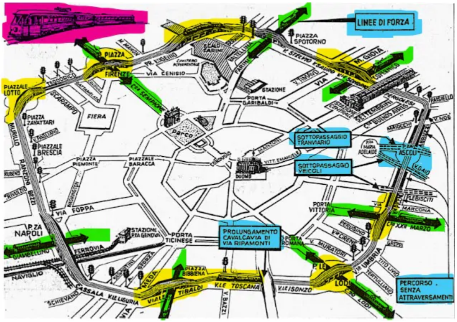


Figure: Ring-Road of a Metropolitan City [5]

- Training Scenarios
- $+\,5\times$ Random Demand
- Each entry of the four (EW, WE, NS, SN) is random
- Demand is in range $[0.3, 0.8]$
- The demand's percentage describes the amount of vehicles per unit of capacity

The FedAvg Algorithm

Listing 1: FedAvg Algorithm

```
def Coordinator():
  θ = Policy.New()
  ∀c ∈ C | SendToClient(c, θ)
  for e ∈ {1,.., E}:
    {(θ_1, w_1)...} = {GetFromClient(c) | ∀c ∈ C}
    {α_1...} = {w_1/Σw_i ...}
    θ = Σα_i θ_i
    ∀c ∈ C | SendToClient(c, θ)
  return θ

def Client(scenario):
  for e ∈ {1,.., E}:
    θ = GetFromCoordinator()
    θ', w = Simulation(θ, scenario)
    SendToCoordinator(θ')
```

- Knowledge is weighted on the complexity of the scenario of a client
- ... which is equal to the demand that the client has to deal with
- Due to resource constrains, only 5 Peers were used, with no client sampling
- The duration of a simulation is $3600s = 1hr = 360$ steps.

Results / Training

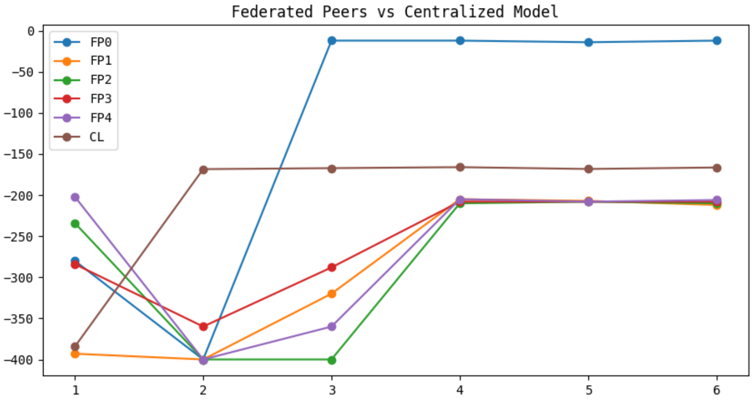


Figure: Total Reward during Training over Time (Episodes)

Results / Evaluation

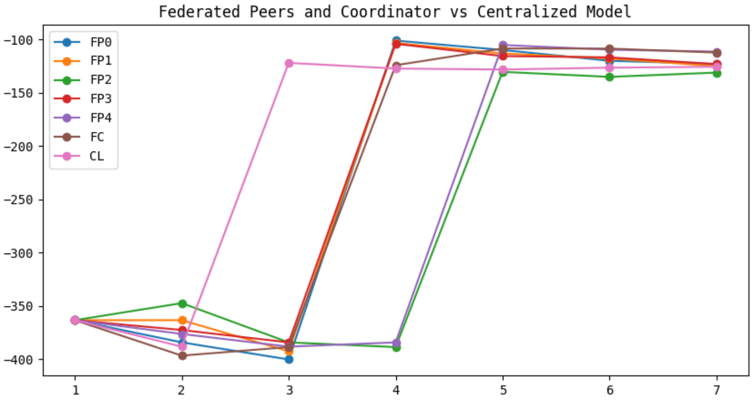


Figure: Average Total Reward during Evaluation over Time (Initial + Episodes)

Results / The Federated Coordinator is Catching Up!

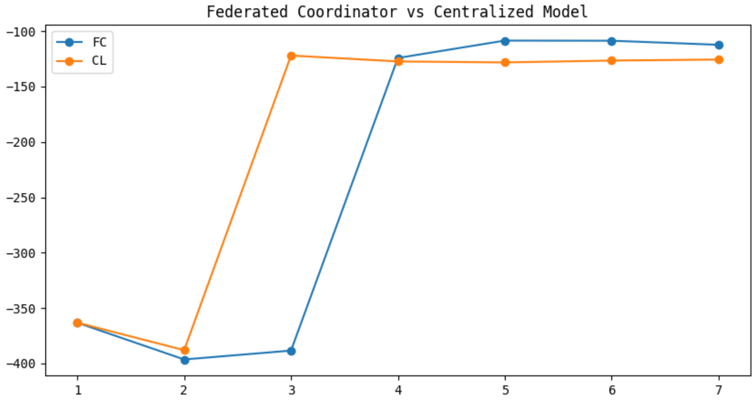


Figure: Average Total Reward during Evaluation over Time (Initial + Episodes)

Comparisons / Fixed-Cycle Agents

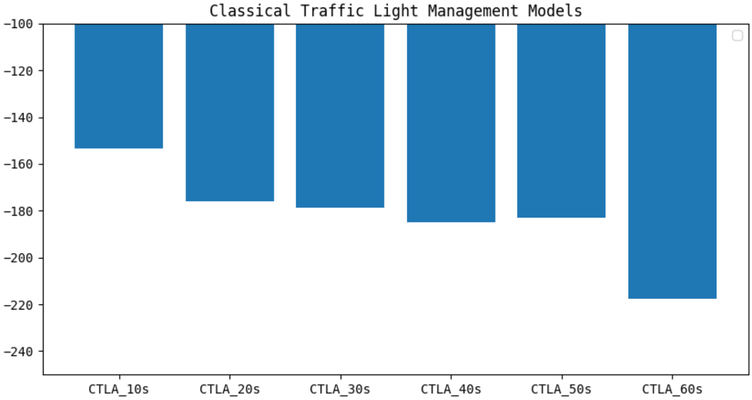


Figure: Average Total Reward during Evaluation of Fixed-Cycle Agents ($CTLA\_Ts$ means T seconds per phase)

Comparisons / Is Deep Q-Learning Worth the Effort?

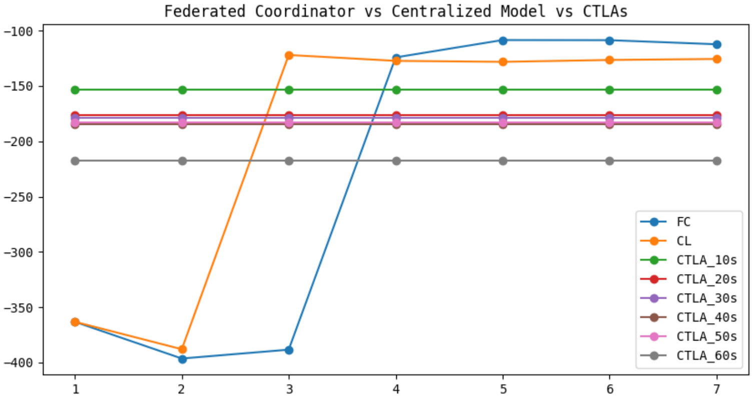
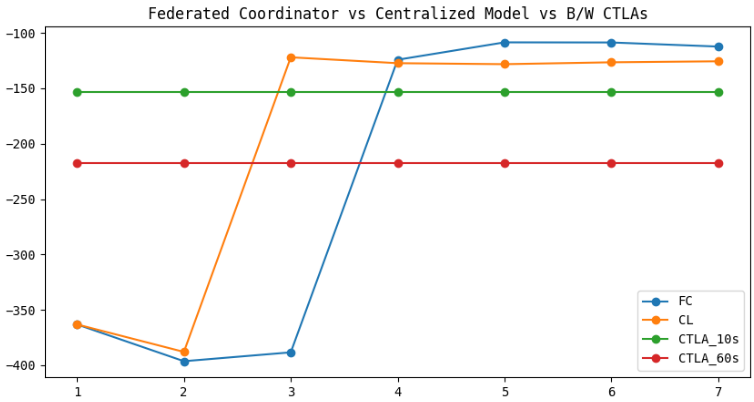


Figure: Comparison of Average Total Reward obtained by Deep Q-Learning Models and Fixed-Cycle Agents during Evaluation

Comparisons / Is Deep Q-Learning Worth the Effort?



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Visualizing that Reward Difference

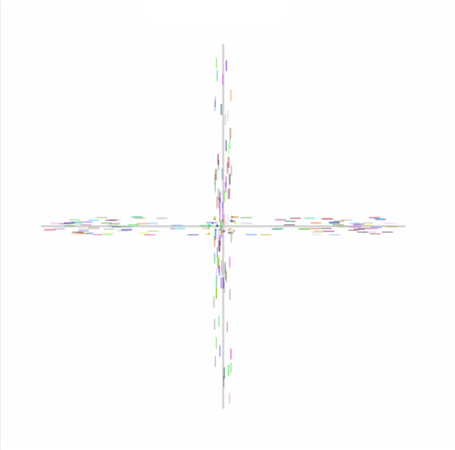


Figure: Microscopic visualization of Fixed-Agent (10s per phase)

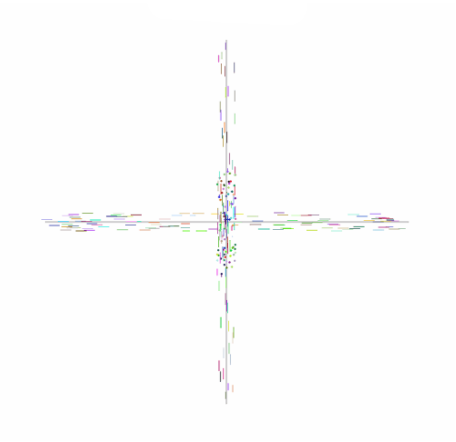


Figure: Microscopic visualization of Fixed-Agent (60s per phase)

Visualizing that Reward Difference

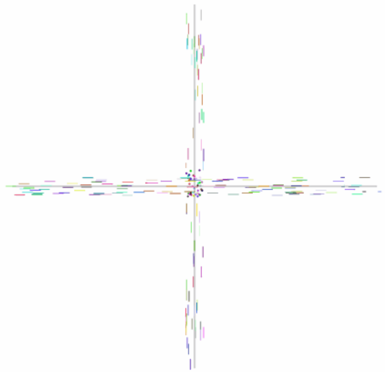


Figure: Microscopic visualization of Centralized Learning Agent

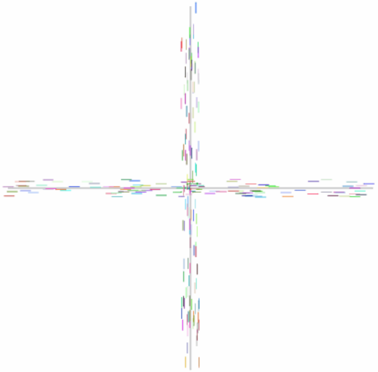
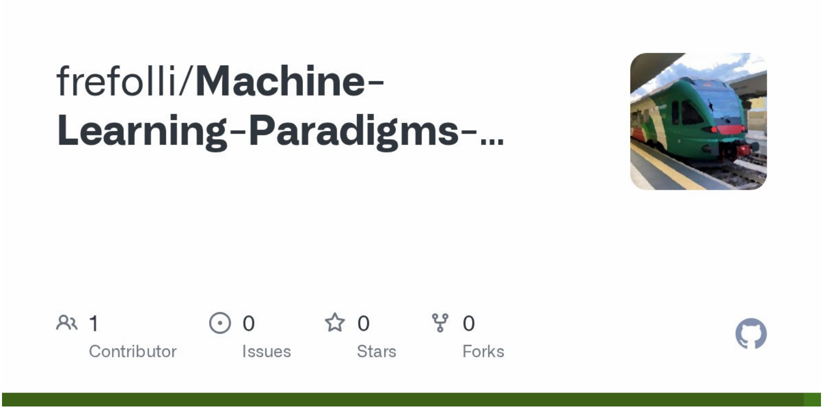


Figure: Microscopic visualization of Federated Learning Agent

Thank You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Bibliography I

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[2] Refolli Francesco. "Unconventional Reinforcement Learning on Traffic Lights with Sumo". In: September 2025. URL: https://github.com/frefolli/master-presentation.

[3] Toru Seo. "UXsim: lightweight mesoscopic traffic flow simulator in pure Python". In: *Journal of Open Source Software* 10.106 (2025), p. 7617. DOI: 10.21105/joss.07617. URL: https://doi.org/10.21105/joss.07617.

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[5] Corriere della Sera. *Disegno del percorso della linea tranviaria circolare per Milano*. 1973.